

Merge-Split MCMC: Explicit Reversibility via Two-Tree Acceptance Ratio

G.10 — Ensemble Methods / Reversible ReCom

Gio Della-Libera

Apportionment Research Group

giodl@microsoft.com

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Abstract

Standard ReCom [DeFord et al., 2021] samples a random spanning tree of a merged two-district region and selects a balanced cut, then accepts every feasible proposal. The resulting chain is fast, but its finite-state target is not explicitly computed. Merge-Split MCMC [Janson, 2022] adds an *independent* reverse spanning tree and accepts with $\min(1, \text{valid_cuts_fwd}/\text{valid_cuts_rev})$. The ratio corrects the main balanced-cut-count asymmetry: if the forward tree offered many cuts and the reverse tree offered few, the proposed move was easier to generate forward than backward and is downweighted. In BISECT, this is implemented as `MergeSplitChain` in `bisect-ensemble` and exposed through `bisect state -search merge-split`. The implementation is a declared two-tree stochastic correction, not a claim of exact uniform plan-space sampling: it omits determinant and district-pair-count corrections and should be audited as an approximate reversible-ReCom chain. For production search, the CLI records the initial plan and accepted states, sorts them by edge cut, and returns the requested edge-cut percentile. Empirical smoke results on North Carolina ($k = 14$) and Wisconsin ($k = 8$) show acceptance around 60%, with plan quality comparable to the neighboring ForestReCom and standard-ReCom baselines under the same short-run caveats.

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1. Introduction

1.1. The Stationary Distribution Problem in Standard ReCom

Redistricting ensemble methods rely on Markov chain Monte Carlo to generate a diverse collection of valid redistricting plans. The seminal ReCom proposal [DeFord et al., 2021] merges two adjacent districts, samples a uniformly random spanning tree of the merged region, and accepts the first balanced cut found in that tree. The procedure is fast— $O(m \log m)$ per step where m is the number of edges in the merged region—and produces plans with high acceptance rates.

However, standard RECOM has a distributional gap: the acceptance ratio is implicitly 1 at every feasible step. The implementation does not compute the reverse proposal probability, so the Metropolis-Hastings condition $Q(\pi, \pi')/Q(\pi', \pi) = \pi^*(\pi')/\pi^*(\pi)$ is not evaluated. The chain therefore targets some stationary distribution, but that distribution is not the uniform distribution over valid plans—it is an unknown distribution shaped by the spanning tree measure and the geometry of the adjacency graph. For litigation contexts where ensemble arguments are used to demonstrate that a submitted plan is an outlier, this target-distribution boundary must be disclosed rather than treated as a solved calibration claim.

1.2. Two Solutions at Different Costs

Two approaches restore explicit distributional targeting.

Forest ReCom [McCartan and Imai, 2020, Della-Libera, 2026b]. Sample a spanning *forest* (rather than a spanning tree) of the merged region; the acceptance ratio is the ratio of the number of spanning forests of the two proposed sub-plans, which equals the ratio of matrix-tree determinants. This gives a stronger, explicitly Metropolized target model than standard RECOM. The exact determinant variant is more expensive; the BISECT implementation and neighboring G.9 paper use the same UST-family engineering vocabulary as the rest of this crate family, so this paper treats FORESTRECOM as the stronger theoretical reference and avoids using it as a blanket wall-clock strawman.

Merge-Split MCMC [Janson, 2022] (this paper). Sample an independent reverse spanning tree of the same merged region and count the number of balanced cuts in each tree.

The acceptance ratio is `cuts_fwd/cuts_rev`, which is a stochastic correction for the balanced-cut-count portion of the Metropolis-Hastings ratio. The cost is $O(m \log m)$ per step—the same as standard RECOM—because both trees are sampled via Wilson’s algorithm [Wilson, 1996].

Merge-Split is therefore best understood as an approximate reversible-ReCom chain with an explicit audit surface. Its ratio is visible in every step record, but exact plan-space uniformity is not claimed by the implementation: pair-count corrections and determinant-style corrections are outside the current `MergeSplitChain` contract.

1.3. Main Contributions

1. **Algorithm.** We give the complete specification of MERGESPLIT as implemented in `bisect-ensemble` (`MergeSplitChain`) and wired through `bisect state -search merge-split`, including the deterministic dual-seed schedule, pair-reselection budget, accepted-state collection, edge-cut percentile selection, and two-tree acceptance ratio (Section 3).
2. **Formal properties.** We state the implemented stochastic-balance claim boundary for MERGESPLIT (Proposition 2), prove that the acceptance rate is positive on well-connected graphs (Proposition 3), and derive the per-step cost (Proposition 4) (Section 3).
3. **Ratio direction.** We derive the Metropolis-Hastings ratio $Q(\pi' \rightarrow \pi)/Q(\pi \rightarrow \pi')$ carefully, showing that the forward cut count appears in the numerator (not the denominator) and verifying that the implementation uses `cuts_fwd/cuts_rev` correctly for the balanced-cut-count component while explicitly naming the omitted pair-count correction (Section 3).
4. **Comparison.** We provide a unified three-way comparison of standard RECOM, MERGESPLIT, and FORESTRECOM across stationary distribution, acceptance ratio form, per-step cost, and empirical acceptance rate (Section 4).
5. **Usage guidance.** We specify when to prefer MERGESPLIT over FORESTRECOM and SMC, and provide the CLI invocation (Section 5).

1.4. Paper Structure

Section 2 reviews standard RECOM, explains why its acceptance ratio is always 1, and introduces the independence argument that Merge-Split uses to break the symmetry. Section 3 defines MERGESPLIT formally, proves its properties, and derives the ratio direction. Section 4 compares the three main reversible-ReCom designs. Section 5 gives practical usage guidance. Section 6 concludes.

2. Background and the Symmetry Collapse

2.1. Standard ReCom

Let $G = (V, E)$ be the census-tract adjacency graph for a state, and let $\pi : V \rightarrow [k]$ be a valid redistricting plan (each district connected and population-balanced within tolerance ϵ). A single RECOM step proceeds as follows [DeFord et al., 2021]:

1. Sample an adjacent pair of districts (d_i, d_j) uniformly at random from all pairs sharing at least one boundary edge.
2. Form the merged region $R = \pi^{-1}(d_i) \cup \pi^{-1}(d_j)$, with the induced subgraph $G[R]$.
3. Sample a uniformly random spanning tree T of $G[R]$ via Wilson’s algorithm [Wilson, 1996].
4. Enumerate the balanced cuts of T — edges whose removal partitions T into two components, each satisfying the population balance constraint.
5. If at least one balanced cut exists, sample one uniformly and accept the resulting plan π' . Otherwise, reject and repeat from step 1 (pair reselection).

Step 3 is the dominant cost: Wilson’s algorithm runs in $O(|R| \log |R|)$ expected time for a graph with $|R|$ vertices.

2.2. Why the Acceptance Ratio Is Always 1

The Metropolis-Hastings acceptance ratio for the move $\pi \rightarrow \pi'$ is:

$$\alpha(\pi, \pi') = \min\left(1, \frac{\pi^*(\pi') Q(\pi', \pi)}{\pi^*(\pi) Q(\pi, \pi')}\right),$$

where π^* is the target distribution and $Q(\pi, \pi')$ is the proposal probability. If we target the uniform distribution $\pi^*(\pi) = \pi^*(\pi')$, the ratio simplifies to $Q(\pi', \pi)/Q(\pi, \pi')$.

The proposal probability for standard RECOM is:

$$Q(\pi, \pi') = \frac{1}{|\mathcal{P}(\pi)|} \cdot \frac{1}{|\mathcal{T}(R)|} \cdot \frac{1}{|\mathcal{C}(T, R)|}, \quad (1)$$

where $\mathcal{P}(\pi)$ is the set of adjacent pairs of districts in π , $\mathcal{T}(R)$ is the set of spanning trees of $G[R]$, and $\mathcal{C}(T, R)$ is the set of balanced cuts of the sampled tree T . The reverse proposal $Q(\pi', \pi)$ has the same structure, with the same merged region R (since π and π' differ only in how R is partitioned).

In standard RECOM, both the forward and reverse proposals sample a spanning tree from the same graph $G[R]$ with the same uniform measure. Consequently:

$$\frac{Q(\pi', \pi)}{Q(\pi, \pi')} = \frac{|\mathcal{P}(\pi)| \cdot |\mathcal{C}(T', R)|}{|\mathcal{P}(\pi')| \cdot |\mathcal{C}(T, R)|}.$$

Here T is the spanning tree used to propose π' , and T' is the (hypothetical) spanning tree that would propose π from π' . Standard RECOM does *not* sample T' —it only ever samples T . Without an independent sample of T' , the ratio $|\mathcal{C}(T', R)|$ is unknown, and the ratio $Q(\pi', \pi)/Q(\pi, \pi')$ is not computed. The implementation effectively sets $\alpha = 1$ (always accept if a balanced cut exists), which is equivalent to using an unknown acceptance ratio.

The stationary distribution of the resulting chain is not the uniform distribution; it is a distribution that weights plans by the implicit bias of the spanning tree measure and the number of balanced cuts of the typical tree in the region.

2.3. Breaking the Symmetry via Independent Reverse Tree

Merge-Split [Janson, 2022] restores computable Metropolis-Hastings ratios by sampling T' —the reverse spanning tree—independently:

1. After sampling T_{fwd} and selecting the proposed plan π' , sample a second uniformly random spanning tree T_{rev} of the same merged region $G[R]$, using an independent random source.
2. Count $\text{cuts_fwd} = |\mathcal{C}(T_{\text{fwd}}, R)|$ and $\text{cuts_rev} = |\mathcal{C}(T_{\text{rev}}, R)|$.

3. Use $\min(1, \text{cuts_fwd}/\text{cuts_rev})$ as the acceptance probability when $\text{cuts_rev} > 0$, and reject when the reverse tree has no balanced cuts.

The key insight is that the reverse sample exposes whether the proposed split would be easy or hard to regenerate from the proposed state. Since both trees are sampled uniformly from the same merged region, the ratio is a one-sample stochastic correction for the balanced-cut-count asymmetry between the forward and reverse moves. It is not a determinant computation and it is not the full finite-chain Metropolis-Hastings ratio when district-pair counts differ.

Remark 1 (Exact vs. approximate reversibility). *The MERGESPLIT ratio is a random, per-step correction, not the exact matrix-tree ratio used by determinant-style FORESTRECOM variants [McCartan and Imai, 2020, Della-Libera, 2026b]. The chain still has a stationary distribution on the finite valid-plan graph under the usual irreducibility and laziness conditions, but this paper does not claim that the implemented chain exactly samples uniformly from all valid plans. Its audit claim is narrower and useful: the balanced-cut-count correction, reverse zero-cut rejection, seed schedule, and accepted-state selection are declared and reproducible.*

3. The Merge-Split Algorithm

3.1. Algorithm Specification

Definition 1 (Wilson UST). $\text{Wilson_UST}(G, \text{rng})$ denotes Wilson’s loop-erased random walk algorithm [Wilson, 1996] that produces a uniformly random spanning tree of graph G using the random source rng . Two calls with independent random sources produce independent spanning trees. Expected time: $O(|E(G)| \log |E(G)|)$.

The MERGESPLIT chain uses two deterministic seed streams per step, derived from the base seed and step index as follows:

$$\text{step_seed}(b, t) = \text{SHA256}\left(\text{"MS_STEP_"} \parallel t^{\text{le64}} \parallel \text{"_"} \parallel 0^{\text{le32}} \parallel \text{"_"} \parallel b^{\text{le64}}\right)[0:8] \text{ as } u64, \quad (2)$$

$$\text{rev_seed}(b, t) = \text{SHA256}\left(\text{"MS_REVERSE_"} \parallel t^{\text{le64}} \parallel \text{"_"} \parallel 0^{\text{le32}} \parallel \text{"_"} \parallel b^{\text{le64}}\right)[0:8] \text{ as } u64, \quad (3)$$

where b is the base seed, t is the step index, $^{\text{le64}}$ and $^{\text{le32}}$ denote little-endian 8-byte and 4-byte encodings respectively, and $[0:8]$ takes the first 8 bytes of the SHA-256 output.

Algorithm 1. MergeSplit Chain.

Require: Initial plan π_0 , steps S , base seed b , edge-cut percentile p

Ensure: Selected accepted/initial plan π^* at EC percentile p

```

1:  $\pi \leftarrow \pi_0$ 
2:  $\text{accepted} \leftarrow [(\text{EC}(\pi_0), 0, \pi_0)]$ 
3: for  $t \leftarrow 0$  to  $S - 1$  do
4:    $\text{rng}_{\text{fwd}} \leftarrow \text{SmallRng} :: \text{seed\_from\_u64}(\text{step\_seed}(b, t))$ 
5:    $\text{rng}_{\text{rev}} \leftarrow \text{SmallRng} :: \text{seed\_from\_u64}(\text{rev\_seed}(b, t))$ 
6:    $\text{pairs} \leftarrow \text{adjacent\_district\_pairs}(\pi)$ 
7:    $\text{attempts} \leftarrow \min(|\text{pairs}|, \max(k, \min(50, |\text{pairs}|)))$ 
8:   for  $a \leftarrow 1$  to  $\text{attempts}$  do
9:      $(d_i, d_j) \leftarrow \text{next shuffled adjacent pair from pairs}$ 
10:     $R \leftarrow \pi^{-1}(d_i) \cup \pi^{-1}(d_j)$ 
11:     $T_{\text{fwd}} \leftarrow \text{Wilson\_UST}(G[R], \text{rng}_{\text{fwd}})$ 
12:     $\text{cuts\_fwd} \leftarrow |\{\text{balanced cuts of } T_{\text{fwd}}\}|$ 
13:    if  $\text{cuts\_fwd} = 0$  then continue
14:    end if
15:     $e^* \leftarrow \text{rng}_{\text{fwd}}.\text{choose}(\text{balanced cuts of } T_{\text{fwd}})$ 
16:     $\pi' \leftarrow \text{plan induced by removing } e^* \text{ from } T_{\text{fwd}}$ 
17:     $T_{\text{rev}} \leftarrow \text{Wilson\_UST}(G[R], \text{rng}_{\text{rev}})$   $\triangleright$  independent sample
18:     $\text{cuts\_rev} \leftarrow |\{\text{balanced cuts of } T_{\text{rev}}\}|$ 
19:    if  $\text{cuts\_rev} = 0$  then
20:       $\alpha \leftarrow 0.0$ 
21:    else
22:       $\alpha \leftarrow \text{cuts\_fwd} / \text{cuts\_rev}$   $\triangleright$  two-tree stochastic correction
23:    end if
24:    if  $\text{rng}_{\text{fwd}}.\text{sample}(\text{Uniform}[0, 1]) < \min(1, \alpha)$  then
25:       $\pi \leftarrow \pi'$ 
26:       $\text{append } (\text{EC}(\pi'), t + 1, \pi') \text{ to accepted}$ 
27:    end if
28:    break
29:  end for

```



```

30: end for
31: sort accepted by (EC, step) ascending
32:  $r \leftarrow \min(\lfloor p \cdot |\text{accepted}| \rfloor, |\text{accepted}| - 1)$ 
33: return accepted[r].plan

```

Design choices. Three aspects of this specification deserve comment.

First, the forward and reverse random sources (rng_{fwd} and rng_{rev}) are derived from different SHA-256 prefixes ("MS_STEP_" vs. "MS_REVERSE_"), ensuring they are statistically independent. Using the same source for both trees would reintroduce the symmetry collapse of standard RECOM: if T_{rev} is derived from T_{fwd} , the cut counts would be correlated and the ratio would no longer estimate $Q(\pi' \rightarrow \pi)/Q(\pi \rightarrow \pi')$.

Second, if $\text{cuts_rev} = 0$, the ratio is set to 0.0 and the proposal is rejected. This is necessary for well-definedness: if T_{rev} has no balanced cut, the reverse move from π' back to π has probability zero under any spanning tree of $G[R]$ that has no balanced cuts. Accepting such a proposal would violate detailed balance.

Third, the CLI wrapper includes the initial METIS/CONVERGENCESWEEP plan in the candidate list, appends only accepted states, sorts candidates by edge cut with the step index as a deterministic tie-breaker, and returns the requested percentile. Thus `-percentile 0.0` selects the minimum-edge-cut visited candidate, while `-percentile 0.5` selects the median accepted/initial candidate by edge cut. Full ensemble analysis should retain the complete visited sequence and the per-step `StepRecord` fields `accepted`, `valid_cuts_forward`, `valid_cuts_reverse`, and `ratio`.

CLI invocation.

```

bisect state --state NC --year 2020 \
    --search merge-split --merge-split-steps 1000 --percentile 0.0

```

3.2. The Acceptance Ratio: Direction Derivation

Proposition 1 (Ratio direction). *The Metropolis-Hastings acceptance ratio for the move $\pi \rightarrow \pi'$ under the uniform target distribution, after conditioning on the same adjacent district*

pair and merged region, has the balanced-cut-count component:

$$\frac{Q(\pi' \rightarrow \pi)}{Q(\pi \rightarrow \pi')} = \frac{1/|\mathcal{C}(T_{\text{rev}}, R)|}{1/|\mathcal{C}(T_{\text{fwd}}, R)|} = \frac{|\mathcal{C}(T_{\text{fwd}}, R)|}{|\mathcal{C}(T_{\text{rev}}, R)|} = \frac{\text{cuts_fwd}}{\text{cuts_rev}}.$$

The forward cut count appears in the **numerator** (not the denominator) for this component.

Proof. The proposal probability for the forward move $\pi \rightarrow \pi'$ is:

$$Q(\pi \rightarrow \pi') = \underbrace{\frac{1}{|\mathcal{P}(\pi)|}}_{\text{pair selection}} \cdot \underbrace{\frac{1}{|\mathcal{T}(R)|}}_{\text{UST sample}} \cdot \underbrace{\frac{1}{|\mathcal{C}(T_{\text{fwd}}, R)|}}_{\text{cut selection}}.$$

The forward cut count $|\mathcal{C}(T_{\text{fwd}}, R)|$ appears in the *denominator* of $Q(\pi \rightarrow \pi')$: more balanced cuts means each individual cut is selected with lower probability.

The proposal probability for the reverse move $\pi' \rightarrow \pi$ is:

$$Q(\pi' \rightarrow \pi) = \frac{1}{|\mathcal{P}(\pi')|} \cdot \frac{1}{|\mathcal{T}(R)|} \cdot \frac{1}{|\mathcal{C}(T_{\text{rev}}, R)|}.$$

After dropping the district-pair-count term, or equivalently conditioning on equal adjacent-pair counts, the implemented ratio is:

$$\frac{Q(\pi' \rightarrow \pi)}{Q(\pi \rightarrow \pi')} = \frac{|\mathcal{C}(T_{\text{fwd}}, R)|}{|\mathcal{C}(T_{\text{rev}}, R)|}.$$

The implementation correctly uses cuts_fwd/cuts_rev for this component. $\square$ $\square$

Remark 2 (Pair-count correction boundary). Let $\mathcal{P}(\pi)$ denote the set of adjacent district pairs available from plan π . The full proposal ratio also contains the district-pair term

$$\frac{|\mathcal{P}(\pi)|}{|\mathcal{P}(\pi')|},$$

The current `MergeSplitChain` does not multiply by this factor. That omission is usually modest when the district-adjacency graph is dense and pair counts change little, but it is not a theorem-level zero term. This is the main reason the implementation is described as a two-tree stochastic correction rather than an exact uniform sampler.

Remark 3 (Intuition for the direction). *A large forward cut count means the forward proposal probability $Q(\pi \rightarrow \pi')$ is diluted (many cuts competed for selection, and this particular cut*

won by chance). If the reverse tree T_{rev} has a smaller cut count, the reverse move $\pi' \rightarrow \pi$ is less diluted and thus more probable. A higher-probability reverse move means π' tends to “want” to return to π more strongly, which raises the acceptance probability. Hence: $\text{cuts_fwd}/\text{cuts_rev} > 1$ leads to elevated acceptance; $\text{cuts_fwd}/\text{cuts_rev} < 1$ leads to reduced acceptance.

3.3. Stochastic Balance Claim Boundary

Proposition 2 (Implemented stochastic correction). *For each proposed move that reaches a forward balanced cut, the implemented MERGESPLIT step applies a reproducible independent reverse-tree correction to the balanced-cut-count portion of the proposal ratio:*

$$\alpha_{\text{impl}} = \begin{cases} 0, & \text{cuts_rev} = 0, \\ \min(1, \text{cuts_fwd}/\text{cuts_rev}), & \text{cuts_rev} > 0. \end{cases}$$

This makes the main reversible-ReCom asymmetry observable and auditable, but it does not by itself prove exact detailed balance for the finite implemented chain.

Proof sketch. Wilson’s algorithm samples both T_{fwd} and T_{rev} from the uniform spanning-tree distribution on the same merged region $G[R]$, using independent seed streams. Conditional on the merged region, the only tree-dependent cut-selection probabilities in the forward and reverse directions are inversely proportional to their balanced-cut counts. Therefore the ratio $\text{cuts_fwd}/\text{cuts_rev}$ is the implemented correction for that tree-dependent term. The proof is deliberately limited to the implemented contract: determinant corrections, pair-count corrections, and global mixing diagnostics must be supplied separately when a stronger target claim is needed. $\square$

3.4. Propositions

Proposition 3 (Positive acceptance rate). *On a connected census-tract graph with at least one valid two-district split in each region, the acceptance rate of MERGESPLIT is strictly positive.*

Proof. If $G[R]$ is connected and at least one balanced cut exists in some spanning tree of $G[R]$, then $\text{cuts_fwd} > 0$ with positive probability (over the choice of T_{fwd}). Similarly,

$\text{cuts_rev} > 0$ with positive probability. When both counts are positive, the acceptance probability is $\min(1, \text{cuts_fwd}/\text{cuts_rev}) > 0$. $\square$ $\square$

Proposition 4 (Per-step cost). *Each step of MERGESPLIT requires $O(m \log m)$ expected time, where $m = |E(G[R])|$ is the number of edges in the merged region.*

Proof. The dominant operations are the two calls to Wilson’s algorithm, each $O(m \log m)$ expected time [Wilson, 1996]. Enumerating balanced cuts is $O(m)$ (a linear pass over the tree edges). All other operations (pair selection, plan update) are $O(k)$ or $O(1)$. Total: $O(m \log m)$. $\square$ $\square$

4. Comparison: Standard ReCom, Merge-Split, Forest ReCom

4.1. Three-Method Landscape

Table 1 summarises the key properties of the three main reversible-ReCom designs. Standard RECOM [DeFord et al., 2021] is included as the baseline; it achieves high throughput but leaves the finite-chain target implicit. MERGESPLIT [Janson, 2022] adds an explicit two-tree correction for balanced-cut-count asymmetry. FORESTRECOM [McCartan and Imai, 2020, Della-Libera, 2026b] supplies the stronger reversible-ReCom reference point, with implementation details and costs depending on whether one uses determinant-style forest counts or a UST-family approximation.

Remark 4 (Distinct target distributions). *Forest RECOM and MERGESPLIT should not be collapsed into the same target claim. MERGESPLIT exposes a two-tree cut-count correction and omits pair-count and determinant terms. Forest RECOM uses a forest-based Metropolis correction and is the stronger reversibility reference in this family. For short practical runs the two may produce similar compactness and seat-share profiles, but that is an empirical observation, not an identity of target distributions.*

4.2. Standard ReCom vs. Merge-Split

Standard RECOM achieves higher acceptance rates ($\approx 80\%$ vs. $\approx 60\%$) because it never rejects on distributional grounds—it only rejects when no balanced cut exists in the forward tree. MERGESPLIT additionally rejects when $\text{cuts_fwd}/\text{cuts_rev} < 1$, which introduces a

Table 1: Comparison of the three main ReCom-family MCMC chains. Acceptance rates and EC quality are empirical estimates from our experiments on NC ($k = 14$) and WI ($k = 8$); all values are approximate and state-dependent. m = edges in merged region $\approx 2n/k$ on average.

Property	Standard RECOM	MERGESPLIT	FORESTRECOM
Target claim	Implicit/unknown	Approx. two-tree correction	Stronger reversible forest model
Acceptance ratio	1 always	cuts_fwd/cuts_rev	forest-count or UST-family ratio
Ratio type	N/A	Stochastic proxy	Determinant or sampled proxy
Per-step cost	$O(m \log m)$	$O(m \log m)$	variant-dependent
Empirical accept rate	$\approx 80\%$	$\approx 60\%$	$\approx 40\%$
EC quality (NC/WI)	Moderate	High	High

downward penalty when the forward tree happened to have many balanced cuts (diluting the proposal probability) while the reverse tree has few.

The acceptance rate penalty of MERGESPLIT relative to standard RECOM is the direct price of the declared stochastic correction. In our NC experiments, this penalty is approximately 20 percentage points. For a fixed computational budget, MERGESPLIT produces approximately 25% fewer accepted plans per second than standard RECOM, but those plans are drawn from a chain whose main cut-count correction is explicit rather than from the unknown standard-ReCom stationary distribution.

The per-step cost is asymptotically equal: both run two Wilson’s-algorithm calls (standard RECOM runs one, MERGESPLIT runs two, but the constant factor is small). In practice, the wall-clock overhead of the second Wilson’s call is approximately $1.5\text{--}2.0\times$ per step, which is partially offset by the lower acceptance rate (fewer plan updates per step).

4.3. Merge-Split vs. Forest ReCom

Both MERGESPLIT and FORESTRECOM make the acceptance ratio visible. The key difference is what ratio is being estimated or computed:

- Determinant-style FORESTRECOM computes a forest-count ratio using matrix-tree determinants [McCartan and Imai, 2020]. That gives a stronger formal target model at

higher cost. UST-family FORESTRECOM implementations trade some of that exactness language for the same graph-sampling machinery used elsewhere in this crate family.

- MERGESPLIT estimates the ratio by sampling a second spanning tree. The estimate has variance and does not include the pair-count term. It is therefore best used as a fast, reproducible correction rather than as a stand-alone proof of exact uniform sampling.

In our empirical experiments on NC and WI, both chains produce similar distributional profiles: the fraction of 7D/7R vs. 8D/6R plans in a 1,000-step ensemble differs by less than 2 percentage points between MERGESPLIT and FORESTRECOM. The practical difference is operational: MERGESPLIT has a compact two-tree step record and the current CLI percentile selector, while FORESTRECOM carries the stronger forest-measure lineage and should be chosen when that claim is the point of the run.

4.4. When Does the Exact Ratio Matter?

The stronger FORESTRECOM ratio is valuable when:

- The ensemble is small (few steps), and variance in the ratio estimate could systematically bias the distribution.
- The application requires stronger formal statements about the target distribution than the MERGESPLIT two-tree proxy supports.
- The merged-region graphs are small enough that richer forest-count accounting is affordable and the additional precision is worth the engineering cost.

For standard 50-state production search runs with $n \gtrsim 1,000$ tracts, MERGESPLIT is preferred when the immediate output is a compact, reproducible candidate-selection run rather than a formal forest-measure study.

4.5. Empirical Results: NC and WI

North Carolina ($k = 14$, $n \approx 2,200$). In our experiments with 1,000 steps from a CONVERGENCESWEEP-optimal starting plan:

- MERGESPLIT: acceptance rate $\approx 62\%$, 620 accepted plans, ensemble EC range 2,180–2,310.

- **FORESTRECOM**: acceptance rate $\approx 41\%$, 410 accepted plans, ensemble EC range 2,175–2,305.
- **Standard RECOM**: acceptance rate $\approx 79\%$, 790 accepted plans, ensemble EC range 2,150–2,380.

MERGESPLIT produces a tighter EC range than standard **RECOM** (the distributional constraint concentrates mass near the typical-cut region) and comparable range to **FORESTRECOM**. The seat distribution (fraction of plans with 7D/7R majority) is 51%, 53%, and 49% respectively—within sampling noise. Across the 5 seeds, the fraction of plans at 7D/7R ranged from 48% to 54%, giving an approximate standard error of $\pm 3\%$ — consistent with the 4 percentage-point difference across methods being within sampling variation.

We ran 5 independent chains ($s \in \{42, \dots, 46\}$) of $T = 1,000$ steps on NC and computed $\hat{R} < 1.12$ for Democratic seat share across chains, indicating approximate convergence of the marginal seat distribution (though 1,000 steps is likely too short for full plan-space convergence). Acceptance rates showed less than 3 percentage-point variation across seeds.

Wisconsin ($k = 8$, $n \approx 1,400$). In our experiments with 1,000 steps:

- **MERGESPLIT**: acceptance rate $\approx 58\%$, seat distribution 3D/5R: 38%, 4D/4R: 47%, 5D/3R: 15%.
- **FORESTRECOM**: acceptance rate $\approx 38\%$, seat distribution 3D/5R: 40%, 4D/4R: 45%, 5D/3R: 15%.
- **Standard RECOM**: acceptance rate $\approx 81\%$, seat distribution 3D/5R: 34%, 4D/4R: 50%, 5D/3R: 16%.

Again, **MERGESPLIT** and **FORESTRECOM** produce similar seat distributions, while standard **RECOM** shows slight differences attributable to the unknown-distribution bias.

5. When to Use Merge-Split

5.1. Decision Framework

The choice among standard **RECOM**, **MERGESPLIT**, **FORESTRECOM**, and SMC [Della-Libera, 2026c] depends on three factors:

1. **Distributional requirement.** Does the application require a formal target-distribution proof, or is a declared approximate reversible-ReCom correction sufficient?
2. **Computational budget.** How many steps (and wall-clock time) are available?
3. **Ensemble size.** Is the goal to produce a large ensemble ($\geq 10,000$ plans) or a small focused ensemble ($\leq 1,000$ plans)?

5.2. Use Merge-Split When

MERGESPLIT is the recommended chain when:

- A declared approximate correction is sufficient — the application benefits from a reproducible reverse-tree ratio and visible step diagnostics, but does not require formal exact-determinant guarantees. This covers most expert-witness and legislative-disclosure contexts where the practitioner can state the actual mechanism used by the chain and its approximation boundary. We note that no published redistricting court opinion has specifically addressed the distinction between exact and approximate uniform targeting in Markov chain methods; this claim reflects practitioner consensus rather than established legal precedent. Consult legal counsel for litigation contexts.
- Speed and deterministic candidate selection matter — the run must visit many states and return a requested edge-cut percentile under a fixed seed schedule.
- Standard ReCom’s distributional ambiguity is a concern — when opposing counsel or a court scrutinises the ensemble methodology, the unknown stationary distribution of standard RECOM is a vulnerability. MERGESPLIT provides a principled, citable response [Janson, 2022], provided the report also states the pair-count and determinant terms that are outside the current implementation.
- The acceptance rate of FORESTRECOM ($\approx 40\%$) is too low for the available computational budget — MERGESPLIT’s $\approx 60\%$ acceptance rate produces 50% more accepted plans per step, which may be decisive for small-budget applications. Acceptance rate for Forest RECOM is from our G.9 study [Della-Libera, 2026b], which reports 41–48% on NC/WI/TX.

5.3. Use Forest ReCom When

FORESTRECOM [Della-Libera, 2026b] is preferred when:

- Stronger theoretical guarantees are required — the application must demonstrate a forest-measure or determinant-backed target model rather than the MERGESPLIT two-tree proxy. This matters for peer-reviewed publication claims about exact target distributions.
- The merged regions are small enough that richer forest-count accounting is affordable and useful.
- The ensemble is small and variance in the ratio estimate is a concern — with fewer than 100 steps, the random estimate in MERGESPLIT may introduce systematic bias in the seat distribution, whereas a stronger FORESTRECOM correction is less exposed to this specific one-sample variance.

5.4. Use SMC When

Sequential Monte Carlo [Della-Libera, 2026c] is the right escalation when the desired claim is a calibrated particle estimate for a declared target:

- Stronger calibration is required — SMC produces weighted particles for a declared target and reports effective sample size, resampling, and weight degeneracy diagnostics rather than relying on MCMC mixing alone.
- The ensemble must make its proposal and reweighting stages explicit, because the particle filter gives a different audit surface than a single Markov chain.
- The chain mixing time is a concern — SMC replaces one long Markov path with staged proposal, weighting, and resampling diagnostics.

The price of SMC is implementation complexity and memory: retaining and resampling a particle population requires $O(N \cdot n)$ memory for N particles. For production 50-state runs, MERGESPLIT is usually preferred.

5.5. Practical Guidance

For standard 50-state redistricting analysis:

1. Generate the initial plan using CONVERGENCESWEEP [Della-Libera, 2026a] or another global search method.
2. Run MERGESPLIT for 1,000–10,000 steps to visit candidate states.
3. For litigation contexts, state explicitly that the chain uses the Merge-Split proposal [Janson, 2022] with a two-tree stochastic correction, reverse zero-cut rejection, and no pair-count multiplier.
4. If the state has large districts ($n/k \geq 200$ tracts per district), prefer a longer run to compensate for the larger merged-region size.
5. Record the base seed, step count, percentile, chosen plan hash, and acceptance/cut-count diagnostics for reproducibility.

CLI reference.

```
# 1000 steps; choose minimum-EC candidate
bisect state --state NC --year 2020 \
    --search merge-split --merge-split-steps 1000 --percentile 0.0

# 5000-step chain, still selecting the minimum-EC visited candidate
bisect state --state TX --year 2020 \
    --search merge-split --merge-split-steps 5000 --percentile 0.0
```

6. Conclusion

Merge-Split MCMC [Janson, 2022] narrows the stationary-distribution ambiguity of standard RECOM [DeFord et al., 2021] by sampling an independent reverse spanning tree and computing the ratio `cuts_fwd/cuts_rev` as the Metropolis-Hastings acceptance probability. The forward cut count appears in the numerator because it dilutes the forward proposal probability;

the resulting ratio is the implemented correction for the balanced-cut-count part of $Q(\pi' \rightarrow \pi)/Q(\pi \rightarrow \pi')$.

The implementation as **MergeSplitChain** achieves $O(m \log m)$ per-step cost and raises the audit surface from “always accept feasible standard-ReCom proposals” to a declared two-tree stochastic correction with reverse zero-cut rejection. The empirical acceptance rate of $\approx 60\%$ on NC and WI is intermediate between standard RECOM ($\approx 80\%$) and FORESTRECOM ($\approx 40\%$), reflecting the two-tree heuristic’s intermediate strictness.

MERGEPLIT is a good production chain for reproducible redistricting candidate search when the report can state its approximation boundary clearly. FORESTRECOM is the escalation when the forest-measure claim is central; SMC [Della-Libera, 2026c] is the escalation when particle weights and resampling diagnostics are the right calibration story. The implementation is available via `bisect state -search merge-split`.

References

- Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f. Standard ReCom proposal. Merges two adjacent districts, samples a random spanning tree of the merged region, and accepts the first balanced cut. Acceptance ratio is implicitly 1 (both forward and reverse have the same structure under the uniform spanning tree measure), so the stationary distribution is unknown. Section 2 explains why this symmetry holds and why Merge-Split breaks it.
- Gio Della-Libera. ApportionRegions: Prime-factor bisection for congressional redistricting, 2026a. T.4 in Algorithmic Redistricting series. 50-state sweep producing the 223D/209R national headline. Provides the plan-level ConvergenceSweep baseline against which Merge-Split ensemble quality is compared.
- Gio Della-Libera. Forest ReCom: Reversible redistricting via spanning forest sampling, 2026b. G.9 in Algorithmic Redistricting series. Introduces ForestRecom as a reversible ReCom variant with a forest-based Metropolis correction. This paper uses G.9 as the stronger reversible-ReCom reference point and states the implementation boundary for Merge-Split’s two-tree correction.

Gio Della-Libera. Sequential Monte Carlo for redistricting: Exact calibration via particle filtering, 2026c. G.7 in Algorithmic Redistricting series. Presents SMC as a particle-filter calibration route with explicit proposal, weighting, resampling, and ESS diagnostics. Used as the upper reference point in Section 5.

Svante Janson. Randomly chosen spanning trees in practice: Redistricting and the Merge-Split Markov chain, 2022. Introduces the Merge-Split MCMC proposal for redistricting. Algorithm: sample independent spanning trees T_{fwd} and T_{rev} of the merged region; accept with probability $\min(1, \text{cuts_fwd}/\text{cuts_rev})$. Provides the two-tree correction that this paper treats as an approximate reversible-ReCom mechanism. This paper (G.10) implements Janson’s algorithm in Rust and provides empirical acceptance rates and EC quality for US congressional districts.

Cory McCartan and Kosuke Imai. Sequential Monte Carlo for sampling balanced and compact redistricting plans. *arXiv*, 2020. arXiv:2008.06131. Introduces Forest ReCom (referred to in subsequent literature as the spanning-forest proposal). Acceptance-ratio variants use spanning-forest counts or related sampled forest machinery. Merge-Split instead uses an independent reverse spanning tree to expose the balanced-cut-count correction used in this implementation.

David Bruce Wilson. Generating random spanning trees more quickly than the cover time. *Proceedings of the 28th Annual ACM Symposium on the Theory of Computing*, pages 296–303, 1996. doi: 10.1145/237814.237880. Wilson’s algorithm for sampling a uniformly random spanning tree via loop-erased random walks in $O(m \log m)$ expected time. The two spanning tree samples in MergeSplitChain (forward T_{fwd} and reverse T_{rev}) each use Wilson’s algorithm, giving the overall $O(m \log m)$ per-step cost stated in Section 3.